ROSEBANK INTERNATIONAL

ICE TASK 4

SQAT6322

Learning units covered: Learning Units 1–8 (integrated revision and application of software quality and testing concepts).

ICE Task learning outcomes

1. Explain and present the five CMMI maturity levels accurately, using relevant examples from the hospital management system (Question 1.1: 15 marks).
2. Discuss how test automation can improve software quality and reliability, using the hospital management system as a practical example (Question 1.2: 6 marks).
3. Explain the importance of test documentation for traceability, communication, quality assurance and future maintenance (Question 1.3: 4 marks).

Assessment alignment

Learning outcome 1 → Q1.1 | Learning outcome 2 → Q1.2 | Learning outcome 3 → Q1.3. Total: 25 marks.

Read and base your answers on the case study presented below.

Case Study: Mamelodi Regional Hospital Health Platform Launch

Background:

Mint Management Group, a software company in South Africa, is embarking on a journey to develop a comprehensive Hospital Management System (HMS) for the newly expanded Mamelodi Regional Hospital. The aim is to digitalise the patient files to improve efficiency.

As a Software Testing Engineer, you are required to evaluate the testing steps for this system to ensure it meets healthcare quality and compliance standards before deployment.

Project Overview:

The health platform includes modules for patient registration, appointment scheduling, electronic medical records (EMR), pharmacy inventory, billing, and reporting. The hospital expects seamless integration, security of patient data, and uninterrupted service delivery once the system goes live.

Instructions:

Answer the questions that follow using the theoretical knowledge gained from this module and relate each answer to the case study.

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Question 1	(Marks: 25)

Q.1.1

Create an infographic illustrating the CMMI model. Include a description of each maturity level and an example that is relevant.

You may use any tool of your choice to create the infographic, as long as you can embed it in the document you submit for the ICE Task.

Description of each maturity level and relevant example	10
The overall appearance of the infographic	5
TOTAL	**15**

(15)

Q.1.2

Discuss the advantages of test automation in a software system, using the case study to illustrate your points.

(6)

Q.1.3

Discuss the importance of test documentation in the software testing process.

(4)